SECURITY ARCHITECTURE & SYSTEM EVALUATION REPORT

CLASSIFICATION: RESTRICTED

Status: **ACTIVE DEPLOYMENT**

KLCDS: Cyber Defense Simulation & Host Protection System

Autonomous Kernel Telemetry, Adversary Kill-Chain Neutralization & Threat Posture Management

Version: 2.4.0 (LTS)
Host Platform: Cross-Platform

EXECUTIVE PURPOSE SUMMARY: The **Kernel-Level Cyber Defense System (KLCDS)** is an advanced cyber defense platform and high-fidelity kill-chain simulation engine. It bridges low-level kernel telemetry (system calls, file system traps, socket state changes) with autonomous, machine-speed countermeasures (process freeze, process termination, dynamic nftables null-routing, decoy canaries, and shadow copy protection). It guarantees continuous system resilience against ransomware, privilege escalation, trojan backdoors, and reconnaissance probes without requiring live destructive malware or bare-metal Linux privileges during validation.

1. System Purpose & Operational Mission

Modern enterprise and defense endpoints face sophisticated threat actors capable of executing multi-stage adversary kill chains in seconds. Traditional signature-only antivirus systems fail against in-memory execution, living-off-the-land (LotL) binaries, and zero-day ransomware. KLCDS was engineered to solve three critical challenges:

- **Deep Kernel-Level Visibility:** Intercepts system events at the syscall boundary (`execve`, `openat`, `setuid`, `rename`, `connect`) to detect malicious intent before disk corruption or unauthorized privilege transitions occur.
- **Autonomous Machine-Speed Containment:** Eliminates human latency in Security Operations Centers (SOC) by automatically triggering kernel emergency freezes (`SIGSTOP`), process kills (`SIGKILL`), and firewall null-routes within milliseconds of a confirmed indicator.
- **Zero-Risk Defensive Validation:** Provides a safe, pure-Python and standards-based simulation engine allowing security engineers to benchmark detection algorithms, train Machine Learning models, and test SIEM pipelines without exposing production networks to operational downtime.

2. System Architecture & Telemetry Pipeline

KLCDS operates on a synchronized four-tier architecture designed for high throughput, minimal overhead, and complete separation between data acquisition, threat scoring, active mitigation, and human oversight:

Layer	Core Components	Key Responsibilities & Operational Capabilities
Presentation & Command	Web Command Center (`index.html`) Rich CLI (`klcds.cli.main`)	Cyberpunk tactical GUI, MITRE ATT&CK; live matrix, Red vs. Blue real-time dual telemetry stream, forensic incident export, and synthetic acoustic synthesizer.

API & Streaming Transport	FastAPI REST (`simulator.api`) WebSocket Streamer (`/ws/stream`)	Asynchronous broadcast of kernel events and alerts at up to 1000 events/sec. Exposes endpoints for SIEM integration, remote trigger, and telemetry replay.
Detection & Scoring Engine	Tripwire Decoys Shannon Entropy ML Isolation Forest Posture Calculator	Calculates real-time Shannon byte entropy (>7.5 triggers ransomware alert), monitors decoy canary access, evaluates process CPU/thread anomalies, and computes composite 0–100 posture score.
Data & Kernel Ingestion	eBPF Probes / Syscall Hooks Journald/Auditd Consumer SQLite Storage (`klcds.db`)	Low-overhead event capture across process execution, file I/O, network sockets, and identity changes. Full relational persistence for forensic auditing.

3. Protective Mechanisms & Active Countermeasures

KLCDS combines passive heuristic tripwires with proactive autonomous intervention to protect systems at every layer:

A. Shannon Entropy Cryptographic Blast Shield

Ransomware encrypts target files using AES or ChaCha20, drastically altering data distribution. Normal documents, source code, and binaries maintain a Shannon entropy score between 3.5 and 5.0. KLCDS samples file write buffers; if consecutive file modifications exceed **7.5 entropy** within a 10-second window, the system flags active encryption.

B. Canary Tripwire Decoys

KLCDS seeds hidden honey-tokens and decoy canary files in `/var/lib/klcds/canary/`. Because legitimate applications have no operational reason to open these files, any `openat` or `unlink` syscall targeting a canary triggers an immediate Critical alert and triggers automatic containment before real user or system directories can be traversed.

C. Privilege Escalation Guard

The system monitors process identity transitions (`setuid`, `setgid`, `capset`). If an unprivileged user process attempts an unauthorized transition to `UID 0` (root) or executes SUID abuse patterns, the engine denies the action, terminates the parent process, and alerts administrators.

D. Dynamic Firewall & Network Containment

Integrated with Linux `nftables` / packet filters, KLCDS monitors connection velocity. Port sweeps and brute-force attempts trigger instantaneous dynamic null-routes against offending IP addresses. Command & Control (C2) callback IPs are instantly banned at the network boundary.

E. Safe Mode vs. Active Enforcement

In default **Safe Mode**, KLCDS logs all containment actions to SQLite without altering production state, allowing operators to verify false-positive rates. When operated with **Active Enforcement (`--enforce`)**, the engine executes real containment: `SIGSTOP` emergency process freeze, `SIGKILL` process tree teardown, file quarantine, and host LAN isolation.

4. Attack Scenarios & Automated Countermeasures

KLCDS includes 7 pre-built adversary scenarios spanning the Cyber Kill Chain and mapped to the MITRE ATT&CK; matrix. The table below details the exact adversary attack vector, detection trigger, and corresponding defensive countermeasure:

Scenario / Vector	MITRE ATT&CK;	Attack Pattern & Trigger	Autonomous Defense Action
Ransomware Mass Encryption	**T1486** (Encrypt for Impact) **T1490** (Inhibit Recovery)	Touches canary honeypot; rapidly renames and encrypts files with >7.9 Shannon entropy; attempts shadow copy purge.	• Canary tripwire alert • `SIGSTOP` emergency freeze • `SIGKILL` process tree purge • LAN network isolation • Shadow backup verification
Trojan Dropper & C2 Shell	**T1204** (User Exec) **T1071** (C2 Protocol) **T1053** (Cron Persistence)	Executes rogue payload; establishes outbound reverse shell; installs crontab persistence for reboot survivability.	• Binary quarantined to `/var/lib/klcds/quarantine` • Crontab persistence purged • Remote C2 IP banned in firewall
SUID Privilege Escalation	**T1068** (Exploitation) **T1548** (Abuse Elevation)	Exploits vulnerability in SUID binary to invoke `setuid(0)` and spawn an unauthorized root shell.	• `setuid(0)` syscall blocked • Parent process terminated • SUID executable capability revoked
SSH Credential Brute-Force	**T1110** (Brute Force) **T1078** (Valid Accounts)	High-velocity dictionary attack exceeding 10 failed PAM authentication events in under 60 seconds.	• PAM account lockout emulation • Dynamic `nftables` null-route blocking source attacker IP
Network Port Reconnaissance	**T1046** (Network Service Scanning)	Rapid horizontal TCP SYN port sweep probing services across the host to map exploitable attack surface.	• Port sweep velocity threshold exceeded • Immediate packet drop rule installed for source IP
Cryptojacking & Resource Drain	**T1496** (Resource Hijacking)	Background mining worker connects to Stratum mining pool and spikes thread utilization.	• Stratum mining handshake detected • ML Isolation Forest flags CPU drain • Worker process terminated
TCP SYN DDoS Flood	**T1498** (Network DoS Flood)	Massive burst of half-open TCP SYN requests designed to exhaust kernel connection backlog.	• Socket backlog monitor alerts • Automated SYN cookies activated to preserve availability

5. Security Posture Scoring & Health Metrics

KLCDS quantifies endpoint security readiness into a real-time **Security Posture Score (0–100%)** calculated across active alerts, unresolved IOCs, and successful countermeasure actions:

POSTURE CALCULATION FORMULA:
Posture Score = Baseline (100) - SUM(Alert_Weight × Severity_Factor) + SUM(Mitigation_Recovery)

- **Critical Alert Penalty:** -25 to -35 points (Ransomware encryption, Root elevation)
- **High Alert Penalty:** -15 to -20 points (C2 Shell, Mass Port Scan)
- **Medium/Low Penalty:** -5 to -10 points (Reconnaissance, Failed Auth)
- **Successful Mitigation Recovery:** +10 to +20 points restored upon verified process kill & IP isolation.

6. Operational Commands & Verification Guide

Security operators and engineers can verify defensive performance via the CLI or the Cyberpunk Web Command Center:

Command	Description & Verification Purpose
`python -m klcds.cli.main sim list`	Enumerate all 7 pre-built attack scenarios with MITRE IDs and severity.
`python -m klcds.cli.main sim run ransomware`	Execute ransomware simulation; triggers canary alarm, entropy calculation, and auto-quarantine.
`python -m klcds.cli.main alerts`	Display audit trail of generated alerts, risk scores (0-100), and timestamps.
`python -m klcds.cli.main posture`	Inspect composite health score and penalty deductions.
`python -m klcds.cli.main web`	Launch the Cyberpunk Command Center (`index.html`) on port 8000.
`python -m pytest tests -v`	Execute 53 automated unit tests validating database, serialization, and scenario execution.

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Verified For: Defense Systems Infrastructure & Cybersecurity Research
Audit Status: VERIFIED & COMPLIANT
Generated Date: October 2026